

Security Audit

Livermore AI (DeFi)

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CAUTION

THIS DOCUMENT IS A SECURITY AUDIT REPORT AND MAY CONTAIN CONFIDENTIAL INFORMATION. THIS INCLUDES IDENTIFIED VULNERABILITIES AND MALICIOUS CODE WHICH COULD BE USED TO COMPROMISE THE PROJECT. THIS DOCUMENT SHOULD ONLY BE FOR INTERNAL USE UNTIL ISSUES ARE RESOLVED. ONCE VULNERABILITIES ARE REMEDIATED, THIS REPORT CAN BE MADE PUBLIC. THE CONTENT OF THIS REPORT IS OWNED BY HASHLOCK PTY LTD FOR USE OF THE CLIENT.

Executive Summary

The Livermore AI team partnered with Hashlock to conduct a security audit of their smart contract. Hashlock manually and proactively reviewed the code in order to ensure the project's team and community that the deployed contracts are secure.

Project Context

The Livermore AI project, featured on the InvestAgentPro website, is an initiative by CitaBlock Technology Corporation aimed at transforming financial market investing through artificial intelligence and Web3 technologies. The goal is to develop an AI agent that learns from the strategies of top global investors and traders, building a data-driven and intelligent investment community. To support this vision, the Livermore token will be launched on the Solana blockchain, with a planned issuance of 210 million tokens over five years. These tokens will fund the platform's technological development and serve as a gateway to financial services such as investment consulting, automated trading, and financing within the Livermore AI ecosystem.

Project Name: Livermore AI

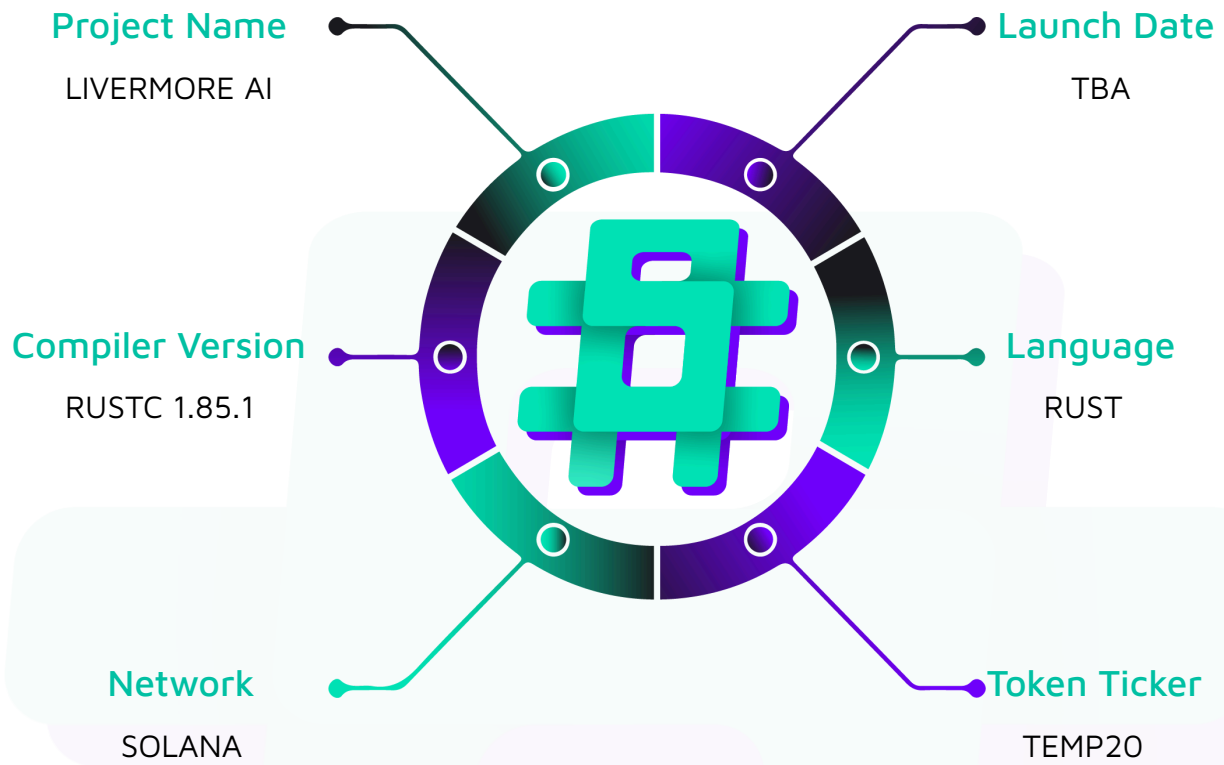
Project Type: Defi, Token, dApp

Compiler Version: rustc 1.85.1

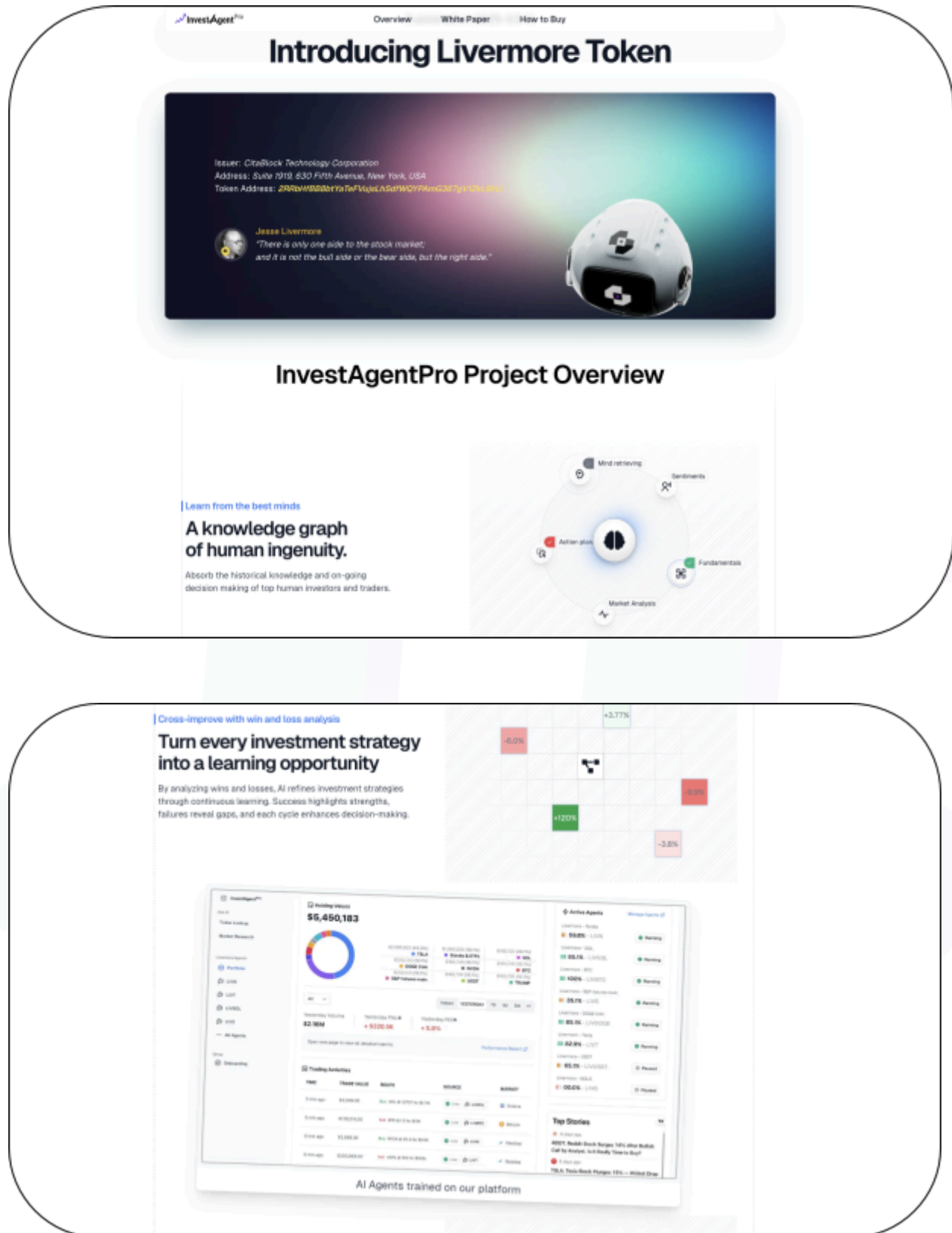
Website: https://www.investagentpro.com/solana_token_launch

Logo:



Visualised Context:

Project Visuals:



Audit scope

We at Hashlock audited the Rust code within the Livermore AI project, the scope of work included a comprehensive review of the smart contracts listed below. We tested the smart contracts to check for their security and efficiency. These tests were undertaken primarily through manual line-by-line analysis and were supported by software-assisted testing.

Description	Livermore AI Smart Contracts
Platform	Solana / Rust
Audit Date	April, 2025
Component	- annual_supply_state.rs - error.rs - instruction.rs - lib.rs - processor.rs - state.rs - Supply_limits.rs - Transfer_hook.rs - utils.rs
GitHub Repo	https://github.com/waylandzhang/livermore-token-vesting
GitHub Commit Hash	1b37d3f9198fc168a0c84a30f2f618ff9a93c256

Security Rating

After Hashlock's Audit, we found the smart contracts to be **"Secure"**. The contracts all follow simple logic, with correct and detailed ordering. They use a series of interfaces.



Not Secure

Vulnerable

Secure

Hashlocked

The 'Hashlocked' rating is reserved for projects that ensure ongoing security via bug bounty programs or on chain monitoring technology.

All issues uncovered during automated and manual analysis were meticulously reviewed and applicable vulnerabilities are presented in the [Audit Findings](#) section. The general security overview is presented in the [Standardised Checks](#) section and the project's contract functionality is presented in the [Intended Smart Contract Functions](#) section.

We initially identified some vulnerabilities that have since been addressed.

Hashlock found:

2 High severity vulnerabilities

2 Medium severity vulnerabilities

1 Low severity vulnerabilities

1 Gas Optimisations

Caution: *Hashlock's audits do not guarantee a project's success or ethics, and are not liable or responsible for security. Always conduct independent research about any project before interacting.*

Intended Smart Contract Functions

Claimed Behaviour	Actual Behaviour
src/instruction.rs - Allows users to: - Create vesting schedules. - Unlock vesting schedules.	Contract achieves this functionality.

Code Quality

This audit scope involves the smart contracts of the Livermore AI project, as outlined in the Audit Scope section. All contracts, libraries, and interfaces mostly follow standard best practices and to help avoid unnecessary complexity that increases the likelihood of exploitation, however, some refactoring was required.

The code is very well commented and closely follows best practice nat-spec styling. All comments are correctly aligned with code functionality.

Audit Resources

We were given the Livermore AI project smart contract code in the form of Github access.

As mentioned above, code parts are well commented. The logic is straightforward, and therefore it is easy to quickly comprehend the programming flow as well as the complex code logic. The comments are helpful in providing an understanding of the protocol's overall architecture.

Dependencies

As per our observation, the libraries used in this smart contracts infrastructure are based on well-known industry standard open source projects.

Apart from libraries, its functions are used in external smart contract calls.

Severity Definitions

The severity levels assigned to findings represent a comprehensive evaluation of both their potential impact and the likelihood of occurrence within the system. These categorizations are established based on Hashlock's professional standards and expertise, incorporating both industry best practices and our discretion as security auditors. This ensures a tailored assessment that reflects the specific context and risk profile of each finding.

Significance	Description
High	High-severity vulnerabilities can result in loss of funds, asset loss, access denial, and other critical issues that will result in the direct loss of funds and control by the owners and community.
Medium	Medium-level difficulties should be solved before deployment, but won't result in loss of funds.
Low	Low-level vulnerabilities are areas that lack best practices that may cause small complications in the future.
Gas	Gas Optimisations, issues, and inefficiencies
QA	Quality Assurance (QA) findings are informational and don't impact functionality. Supports clients improve the clarity, maintainability, or overall structure of the code.

Status Definitions

Each identified security finding is assigned a status that reflects its current stage of remediation or acknowledgment. The status provides clarity on the handling of the issue and ensures transparency in the auditing process. The statuses are as follows:

Significance	Description
Resolved	The identified vulnerability has been fully mitigated either through the implementation of the recommended solution proposed by Hashlock or through an alternative client-provided solution that demonstrably addresses the issue
Acknowledged	The client has formally recognized the vulnerability but has chosen not to address it due to the high cost or complexity of remediation. This status is acceptable for medium and low-severity findings after internal review and agreement. However, all high-severity findings must be resolved without exception.
Unresolved	The finding remains neither remediated nor formally acknowledged by the client, leaving the vulnerability unaddressed.

Audit Findings

High

[H-01] src/instruction.rs#create_vesting_schedules - Incorrect release time for VestingType::Market

Description

In line 310, the `release_time` for the `VestingType::Market` is set to `start_time` and `MONTH_IN_SECONDS`. This is incorrect because, based on the comment in line 308, it should be set to `YEAR_IN_SECONDS`.

Impact

`VestingType::Market` will unlock earlier than intended.

Recommendation

Consider updating the `release_time` to be `start_time` and `YEAR_IN_SECONDS` for `VestingType::Market`.

Status

Resolved

[H-02] `src/processor.rs#process_create` - Funds are stuck if transaction is replayed

Description

In line 135, the vesting schedules are only added to the PDA `vesting_account` if it is empty. This is problematic because if the transaction is dispatched or replayed with the same parameters, the user funds are sent to the ATA in line 300, but the PDA `vesting_account` does not record the newly added funds.

Impact

The user's funds will be lost.

Recommendation

Consider returning an error if the PDA `vesting_account` is not empty.

Status

Resolved

Medium

[M-01] `src/processor.rs#process_create` - Signer validation not enforced

Description

The `source_token_account_owner` is not checked to be the signer of the transaction, which is inconsistent with `src/instruction.rs:225`.

Impact

The transaction can be executed when the `source_token_account_owner` is not the signer.

Recommendation

Consider checking that the `source_token_account_owner` is the signer of the transaction.

Status

Resolved

[M-02] `src/utils.rs#calculate_project_year` - Inconsistent seconds per year definition

Description

The `SECONDS_PER_YEAR` is defined to represent the number of seconds per year. It is computed as $365 * 24 * 60 * 60$.

However, this is inconsistent with the `YEAR_IN_SECONDS` constant, which is defined in `src/instruction.rs:9`. It is computed as $12 * 30 * 24 * 60 * 60$.

Impact

Inconsistent definition of seconds per year throughout the protocol.

Recommendation

Consider using the `YEAR_IN_SECONDS` constant and removing `SECONDS_PER_YEAR`.

Status

Resolved

Low

[L-01] `src/processor.rs#process_create` - Mint token decimals may be different than `TOKEN_DECIMALS`

Description

In line 270, the decimal amount from the mint token is not validated to be the same as `TOKEN_DECIMALS`, which is defined in `src/supply_limits.rs:10`.

Recommendation

Consider adding validation to ensure the mint token decimal is the same as `TOKEN_DECIMALS`.

Status

Resolved

Gas

[G-01] `src/annual_supply_state.rs` - Duplicate validations

Description

The `validate_issuance` function in line 92 calls `validate_yearly_limit` for validation. This is unneeded because it has already been validated previously in `src/processor.rs:66`.

Other than that, the `header.destination_address` validation in `src/processor.rs:510` is duplicate with `src/processor.rs:460`.

Recommendation

Consider removing the `crate::supply_limits::validate_yearly_limit` call and the `header.destination_address` validation in `src/processor.rs:510`.

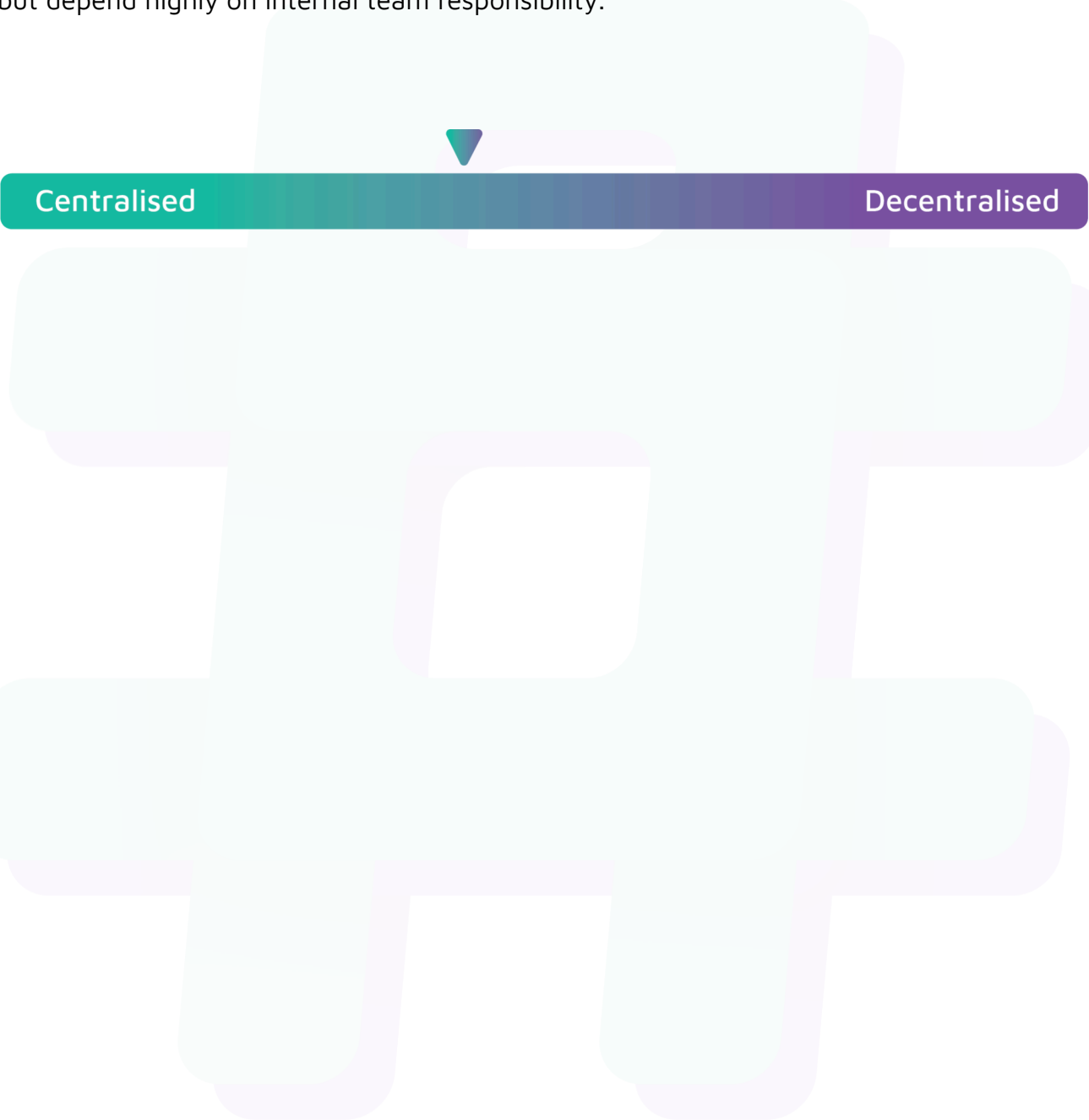
Status

Resolved

Centralisation

The Livermore AI project values security and utility over decentralisation.

The owner executable functions within the protocol increase security and functionality but depend highly on internal team responsibility.



Centralised

Decentralised

Conclusion

After Hashlock's analysis, the Livermore AI project seems to have a sound and well-tested code base, now that our vulnerability findings have been resolved and acknowledged. Overall, most of the code is correctly ordered and follows industry best practices. The code is well commented on as well. To the best of our ability, Hashlock is not able to identify any further vulnerabilities.

Our Methodology

Hashlock strives to maintain a transparent working process and to make our audits a collaborative effort. The objective of our security audits is to improve the quality of systems and upcoming projects we review and to aim for sufficient remediation to help protect users and project leaders. Below is the methodology we use in our security audit process.

Manual Code Review:

In manually analysing all of the code, we seek to find any potential issues with code logic, error handling, protocol and header parsing, cryptographic errors, and random number generators. We also watch for areas where more defensive programming could reduce the risk of future mistakes and speed up future audits. Although our primary focus is on the in-scope code, we examine dependency code and behaviour when it is relevant to a particular line of investigation.

Vulnerability Analysis:

Our methodologies include manual code analysis, user interface interaction, and white box penetration testing. We consider the project's website, specifications, and whitepaper (if available) to attain a high-level understanding of what functionality the smart contract under review contains. We then communicate with the developers and founders to gain insight into their vision for the project. We install and deploy the relevant software, exploring the user interactions and roles. While we do this, we brainstorm threat models and attack surfaces. We read design documentation, review other audit results, search for similar projects, examine source code dependencies, skim open issue tickets, and generally investigate details other than the implementation.

Documenting Results:

We undergo a robust, transparent process for analysing potential security vulnerabilities and seeing them through to successful remediation. When a potential issue is discovered, we immediately create an issue entry for it in this document, even though we have not yet verified the feasibility and impact of the issue. This process is vast because we document our suspicions early even if they are later shown to not represent exploitable vulnerabilities. We generally follow a process of first documenting the suspicion with unresolved questions, and then confirming the issue through code analysis, live experimentation, or automated tests. Code analysis is the most tentative, and we strive to provide test code, log captures, or screenshots demonstrating our confirmation. After this, we analyse the feasibility of an attack in a live system.

Suggested Solutions:

We search for immediate mitigations that live deployments can take and finally, we suggest the requirements for remediation engineering for future releases. The mitigation and remediation recommendations should be scrutinised by the developers and deployment engineers, and successful mitigation and remediation is an ongoing collaborative process after we deliver our report, and before the contract details are made public.

Disclaimers

Hashlock's Disclaimer

Hashlock's team has analysed these smart contracts in accordance with the best industry practices at the date of this report, in relation to: cybersecurity vulnerabilities and issues in the smart contract source code, the details of which are disclosed in this report, (Source Code); the Source Code compilation, deployment, and functionality (performing the intended functions).

Due to the fact that the total number of test cases is unlimited, the audit makes no statements or warranties on the security of the code. It also cannot be considered as a sufficient assessment regarding the utility and safety of the code, bug-free status, or any other statements of the contract. While we have done our best in conducting the analysis and producing this report, it is important to note that you should not rely on this report only. We also suggest conducting a bug bounty program to confirm the high level of security of this smart contract.

Hashlock is not responsible for the safety of any funds and is not in any way liable for the security of the project.

Technical Disclaimer

Smart contracts are deployed and executed on a blockchain platform. The platform, its programming language, and other software related to the smart contract can have their own vulnerabilities that can lead to attacks. Thus, the audit can't guarantee the explicit security of the audited smart contracts.

About Hashlock

Hashlock is an Australian-based company aiming to help facilitate the successful widespread adoption of distributed ledger technology. Our key services all have a focus on security, as well as projects that focus on streamlined adoption in the business sector.

Hashlock is excited to continue to grow its partnerships with developers and other web3-oriented companies to collaborate on secure innovation, helping businesses and decentralised entities alike.

Website: hashlock.com.au

Contact: info@hashlock.com.au



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